ESP32 pinout connection chart

Functional GPIO labels only. Exact DevKit header positions/USB power path remain unverified; no wiring orientation implied.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

GPIO / terminal	Firmware signal	Exact proposed endpoint chain	Electrical status / requirement
GPIO16	L_FWD	U1.2 -> U1.18 -> DRV_L.RPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO17	L_REV	U1.4 -> U1.16 -> DRV_L.LPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO18	R_FWD	U1.6 -> U1.14 -> DRV_R.RPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO19	R_REV	U1.8 -> U1.12 -> DRV_R.LPWM	**BLOCKED_MISSING** U1 + bias parts missing
GPIO14	ENABLE	U1.11 -> U1.9 -> S2 EN NC -> six enables	**BLOCKED_MISSING** U1 + bias + cutoff missing
GPIO32	TF	U1.13 -> U1.7 -> US_F.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO33	TL	U1.15 -> U1.5 -> US_L.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO13	TR	U1.17 -> U1.3 -> US_R.TRIG	**BLOCKED_MISSING** U1 + bias parts missing
GPIO21	BRUSH_IN1	DRV_A.IN1 / both brushes	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO22	BRUSH_IN2	DRV_A.IN2 / held LOW	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO23	CONVEYOR_IN3	DRV_A.IN3 / conveyor	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO25	CONVEYOR_IN4	DRV_A.IN4 / held LOW	**PROVISIONAL** Carrier/pulls unverified; HOLD
GPIO34	EF	ECHO -> R23/R24 -> U2.1/7 -> GPIO34	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO35	EL	ECHO -> R25/R26 -> U2.3/5 -> GPIO35	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO36	ER	ECHO -> R27/R28 -> U2.6/2 -> GPIO36 / VP	**BLOCKED_MISSING** ECHO disconnected; parts missing
GPIO27	ESTOP_OK	3V3 -> S2 NC -> K1 sense NO -> GPIO27	**BLOCKED_MISSING** Cutoff/feedback parts missing

5V/VIN: only V5_MCU after OV1 and actual-board qualification. 3V3: E-stop dry feedback + U2 only. GND: logic return to BUCK1 OUT-.
Never feed 5V into a GPIO or 3V3 pin.

Unused: GPIO26, GPIO39/VN, GPIO4. Reserve GPIO1/3 for serial, GPIO0/2/5/12/15 boot straps, GPIO6-11 flash. Keep EN/reset under onboard control; no JTAG on GPIO13/14. ESP32-CAM remains disconnected.

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.